

Confidence-based position selection

Does choosing *which* position to unmask buy anything?

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1 The idea, and why it is MDLM-only

In absorbing-state diffusion the reverse posterior at a masked position is

$$P(\text{stay masked}) = \frac{\text{move_chance}_s}{\text{move_chance}_t},$$

the *same* value at every masked position and independent of the token distribution. The base sampler’s choice of which position to unmask therefore carries no information at all. That is free capacity, and this report asks whether spending it well is worth anything.

Two structural restrictions follow, and both are properties of the method rather than of the implementation:

- **MDLM only.** Uniform diffusion (UDLM) has no mask state — every position is resampled at every step — so there is no “which position to unmask” to decide. The code raises `NotImplementedError` rather than silently doing nothing.
- **marginal only.** Under `edlm` the per-position distribution is a one-hot on the single drawn candidate, so the confidence is 1 at every masked position and the ranking is entirely ties. Only `marginal`, which averages the N normalised posteriors, exposes a usable per-position signal.

The second point is the one that mattered for the paper’s framing: the $\lambda \times N$ sweep found `marginal` and `edlm` empirically interchangeable, so `marginal` could not be justified on performance. A capability that exists only under `marginal` would have justified it instead. That is what is being tested here.

2 The arms

A `bernoulli` — the schedule’s own coin flip. The original sampler, and the baseline that actually matters.

B `random_k` — deterministic count, uniformly random positions. Isolates fixing the *count*.

C `xtheta_conf` — deterministic count, ranked by the denoiser marginal x_θ . Isolates ranking *at all*; this is MaskGIT-style confidence decoding.

D `xbar_conf` — ranked by the reward-weighted candidate histogram $\bar{x}_0$. The reward-aware version.

E `cv_conf` — D with a control-variate correction (Section 4). No free parameter.

Everything below is MDLM, **marginal**, $N = 300$, 1,024 samples per run, $T = 32$ steps, and **5 seeds** $\times$ **3 values of λ** **15 runs per arm**. Seed replication is not optional here: two runs of an *identical* algorithm differing only in random stream gave hit counts of 30 and 49, so single-seed rows are not interpretable.

3 Result: nothing beats doing nothing

arm	n	novel	novel-QED	hits (≥ 0.6)
A <code>bernoulli</code>	15	362.3 ± 6.0	0.5044 ± 0.0017	30.1 ± 1.7
B <code>random_k</code>	15	345.7 ± 6.7	0.5022 ± 0.0019	27.4 ± 1.4
C <code>xtheta_conf</code>	15	302.7 ± 6.0	0.5055 ± 0.0019	22.6 ± 1.6
D <code>xbar_conf</code>	15	261.5 ± 13.2	0.5110 ± 0.0011	26.1 ± 2.3
E <code>CV</code>	15	282.0 ± 9.2	0.5111 ± 0.0011	27.3 ± 2.3

Table 1: QED: the five arms, mean $\pm$ standard error over 15 runs.

arm	Δ novel	Δ property	Δ hits
B <code>random_k</code>	$-16.6 \pm 4.5^*$	-0.0022 ± 0.0016	-2.7 ± 1.6
C <code>xtheta_conf</code>	$-59.5 \pm 3.7^*$	$+0.0011 \pm 0.0013$	$-7.5 \pm 1.4^*$
D <code>xbar_conf</code>	$-100.7 \pm 9.6^*$	$+0.0066 \pm 0.0015^*$	$-4.0 \pm 1.2^*$
E <code>CV</code>	$-80.3 \pm 5.3^*$	$+0.0067 \pm 0.0012^*$	-2.7 ± 2.1

* = $|t| > 2.145$, i.e. $p < 0.05$ on a paired t -test with 14 degrees of freedom.

Table 2: QED: paired against A `bernoulli`, within each (λ, seed) . Pairing removes the λ and seed effects, which otherwise dominate the spread.

arm	n	novel	novel-ring count	hits (≥ 5)
A <code>bernoulli</code>	15	303.5 ± 5.9	3.2672 ± 0.0211	83.1 ± 2.0
B <code>random_k</code>	15	287.9 ± 4.5	3.3693 ± 0.0259	83.4 ± 1.9
C <code>xtheta_conf</code>	15	256.9 ± 4.2	3.2710 ± 0.0393	72.6 ± 2.3
D <code>xbar_conf</code>	15	187.5 ± 2.4	3.4438 ± 0.0934	60.5 ± 2.7
E <code>CV</code>	15	235.0 ± 2.7	3.3590 ± 0.0685	69.2 ± 2.9

Table 3: ring count: the five arms, mean $\pm$ standard error over 15 runs.

Reading the three tables together:

1. **No arm beats A `bernoulli` on hits.** Two are significantly worse; the best (E) is a draw. Doing nothing is the best policy on the metric a fixed generation budget cares about.

arm		Δ novel	Δ property	Δ hits
B	random_k	$-15.6 \pm 3.6^*$	$+0.1021 \pm 0.0261^*$	$+0.3 \pm 1.9$
C	xtheta_conf	$-46.5 \pm 4.6^*$	$+0.0038 \pm 0.0320$	$-10.5 \pm 1.9^*$
D	xbar_conf	$-115.9 \pm 6.1^*$	$+0.1766 \pm 0.0869$	$-22.6 \pm 2.0^*$
E	CV	$-68.5 \pm 5.4^*$	$+0.0918 \pm 0.0621$	$-13.9 \pm 2.3^*$

* = $|t| > 2.145$, i.e. $p < 0.05$ on a paired t -test with 14 degrees of freedom.

Table 4: ring count: paired against A bernoulli, within each (λ, seed) . Pairing removes the λ and seed effects, which otherwise dominate the spread.

step	isolates	Δ novel	Δ hits
A $\rightarrow$ B	fixing the count	$-16.6 \pm 4.5^*$	-2.7 ± 1.6
B $\rightarrow$ C	ranking at all	$-42.9 \pm 4.3^*$	$-4.8 \pm 1.0^*$
C $\rightarrow$ D	making the ranking reward-aware	$-41.2 \pm 8.9^*$	$+3.5 \pm 1.8$
C $\rightarrow$ E	reward-aware, control-variate corrected	$-20.7 \pm 5.3^*$	$+4.7 \pm 1.6^*$

The damage is done by *ranking*, not by the reward: B $\rightarrow$ C costs the novelty. Making the ranking reward-aware (C $\rightarrow$ E) wins part of it back — the contribution repairs a wound the ranking itself inflicts.

Table 5: QED: the decomposition the four arms were designed for, each step paired.

2. **The cost is ranking itself, not the reward.** Introducing a confidence ranking (B $\rightarrow$ C) is what destroys novelty: a confidence rule commits to the most confident, hence most typical, positions and walks towards the mode of the data. Reward-awareness (C $\rightarrow$ E) then wins some of it back.
3. **One thing does survive.** E raises the property mean significantly and by the largest margin of any arm. If the objective were the mean quality of novel molecules rather than the count above a bar, E wins; on hits it does not.

4 The control variate

$\bar{x}_0$ is a *self-normalised* importance sampling estimate: the weights $w_n = \text{softmax}(\lambda r_n)$ share a denominator over all N samples, so it is a ratio of two averages and is **biased at finite N** , by $O(1/N)$ — measured at 0.104 for $N=5$ falling to 0.0007 for $N=1000$ on a synthetic check. It is consistent, not unbiased.

The same N candidates averaged with *uniform* weights, $\bar{x}_0^{(0)}$, has $\mathbb{E}[\bar{x}_0^{(0)}] = x_\theta$ *exactly*, because the candidates are drawn i.i.d. from x_θ . So $\bar{x}_0^{(0)} - x_\theta$ is a zero-mean quantity computed from the same draws, hence correlated with the error in $\bar{x}_0$:

$$\bar{x}_0^{\text{CV}} = \bar{x}_0 - c(\bar{x}_0^{(0)} - x_\theta).$$

Two things this is and is not. It **adds** no bias — the correction has mean zero for any c — but it does **not remove** the SNIS bias already in $\bar{x}_0$. And it costs no extra reward evaluations, since the candidates were already scored; reusing those same draws is not an economy but the *condition* for the correction to reduce variance rather than add noise.

Classical theory gives $c^* = \text{Cov}/\text{Var}$, the OLS regression coefficient, with variance falling by $(1-\rho^2)$. And $c = 1$ is not arbitrary: at any position the weights do not depend on, the indicator draws are i.i.d. and $\text{Cov} = \text{Var} = p(1-p)/N$ exactly, so $c^* = 1$. Departures from 1 measure how strongly the reward couples to that position.

c	novel	novel-QED	hits	Δ novel vs $c=1$	Δ hits
$c=0$	260.4 ± 13.5	0.5111 ± 0.0011	25.5 ± 2.3	$-21.6 \pm 6.6^*$	-1.9 ± 2.0
$c=0.5$	255.5 ± 9.7	0.5092 ± 0.0011	23.5 ± 1.8	$-26.5 \pm 2.3^*$	-3.8 ± 2.0
$c=1$	282.0 ± 9.2	0.5111 ± 0.0011	27.3 ± 2.3	(reference)	
$c=1.5$	273.5 ± 11.7	0.5092 ± 0.0014	26.0 ± 2.2	-8.5 ± 5.7	-1.3 ± 1.9
$c=2$	278.8 ± 10.3	0.5073 ± 0.0015	24.5 ± 2.0	-3.2 ± 4.4	-2.8 ± 1.9

$c = 1$ is best or tied everywhere and every difference has the same sign, so the theoretical default is also the empirical optimum. An adaptive c estimated from the same samples would re-introduce bias for no measurable gain.

Table 6: QED: sweeping the control-variate coefficient, 15 runs per c . c is not restricted to $[0, 1]$ — Cov/Var can exceed 1.

c	novel	novel-ring count	hits	Δ novel vs $c=1$	Δ hits
$c=0$	186.8 ± 2.4	3.4473 ± 0.0950	60.3 ± 2.7	$-48.2 \pm 2.8^*$	$-8.9 \pm 2.1^*$
$c=0.5$	199.6 ± 3.1	3.3070 ± 0.0573	58.0 ± 1.7	$-35.4 \pm 2.5^*$	$-11.2 \pm 2.2^*$
$c=1$	235.0 ± 2.7	3.3590 ± 0.0685	69.2 ± 2.9	(reference)	
$c=1.5$	218.6 ± 3.7	3.3782 ± 0.0564	66.0 ± 2.3	$-16.4 \pm 2.2^*$	-3.2 ± 2.5
$c=2$	220.9 ± 2.7	3.3997 ± 0.0564	68.2 ± 2.0	$-14.1 \pm 1.8^*$	-1.0 ± 2.6

$c = 1$ is best or tied everywhere and every difference has the same sign, so the theoretical default is also the empirical optimum. An adaptive c estimated from the same samples would re-introduce bias for no measurable gain.

Table 7: ring count: sweeping the control-variate coefficient, 15 runs per c . c is not restricted to $[0, 1]$ — Cov/Var can exceed 1.

5 Is there a critical time window?

The failure mode above suggests a repair. Early unmasking decisions fix the molecular scaffold, which is where novelty is decided; late ones are local substitutions. Confining the ranking to late t should then keep the quality gain without the diversity cost. The reward tilt stays on for the whole trajectory — only the position rule is windowed, through a `position_t_max` that is deliberately separate from the guidance window.

t runs from 1 (noise) to 0 (clean), so $w = 0.25$ means the ranking is active only on the last quarter of the trajectory. $w = 0$ is exactly `bernoulli` and $w = 1$ is exactly arm E — both verified to reproduce those rows.

The hypothesis is refuted, and the way it fails is informative. No w beats `bernoulli` on hits. But novelty and quality do not live in the same part of the trajectory in the way the hypothesis assumed — they live in the *same* part:

- Novelty is lost as soon as the window opens at all: $w = 0.25$ already costs a significant number of novel molecules, while delivering essentially no property gain.

w	ranking active on	novel	novel-QED	hits	Δ novel	Δ prop.	Δ hits
$w=0$	= bernoulli	362.3 ± 6.0	0.5044 ± 0.0017	30.1 ± 1.7	(reference)		
$w=0.25$	last 0.25 of t	346.9 ± 5.3	0.5053 ± 0.0020	29.1 ± 1.6	$-15.3 \pm 4.5^*$	$+0.0009 \pm 0.0014$	-1.0 ± 1.4
$w=0.5$	last 0.5 of t	345.1 ± 6.3	0.5028 ± 0.0020	27.6 ± 2.3	$-17.1 \pm 4.4^*$	-0.0016 ± 0.0012	-2.5 ± 1.9
$w=0.75$	last 0.75 of t	322.6 ± 7.7	0.5061 ± 0.0020	26.8 ± 1.8	$-39.7 \pm 4.6^*$	$+0.0018 \pm 0.0011$	$-3.3 \pm 1.3^*$
$w=1$	= full trajectory (arm E)	282.0 ± 9.2	0.5111 ± 0.0011	27.3 ± 2.3	$-80.3 \pm 5.3^*$	$+0.0067 \pm 0.0012^*$	-2.7 ± 2.1

Table 8: QED: the CV ranking confined to $t \leq w$, paired against $w = 0$.

- The property gain appears only at $w = 1$. It comes from the early, high-noise steps — exactly the steps that cost the most novelty.

So the two effects are produced by the same decisions and cannot be separated by windowing. Late-step ranking is the worst of both: it costs diversity and buys nothing, presumably because by then x_t is nearly determined and the order of the remaining unmaskings barely matters.

6 Implementation checks, and two bugs

Five identities were available for free, and all five hold:

1. $\lambda = 0 \Rightarrow \mathbf{CV} \equiv \mathbf{xtheta_conf}$. With uniform weights $\bar{x}_0 = \bar{x}_0^{(0)}$, so the correction collapses to x_θ algebraically. Measured agreement: 0.0001–0.0010, i.e. floating point.
2. $c = 0 \Rightarrow \mathbf{CV} \equiv \mathbf{xbar_conf}$. Holds within seed noise, though not to floating point — tiny numerical differences flip near-ties in the ranking and trajectories then diverge.
3. $w = 0 \Rightarrow \mathbf{bernoulli}$. Reproduces it exactly, run for run.
4. $\beta = 0 \equiv \mathbf{C}$ and $\beta = 1 \equiv \mathbf{D}$ for the blend, checked on CPU before launching.
5. Under **edlm**, **xbar_conf** degenerates to **random_k**, as the one-hot argument requires.

Check 5 initially *failed*, and finding out why exposed the first bug.

Bug 1 — stable-sort tie-breaking. **argsort** returns ties in index order, so when every masked position scored identically the arm unmasked *left to right* rather than at random: a systematically different sampler. This is invisible under **marginal**, where scores genuinely differ, but it governs the whole **edlm** check. Fixed with a 10^{-6} jitter. The entire four-arm ablation was re-run afterwards, and the conclusions changed: before the fix C looked strong and D looked weak, which was an artifact.

Bug 2 — a falsy default. `float(getattr(..., 'cv_coeff', 1.0) or 1.0)` maps $c = 0$ to $c = 1$, because 0.0 is falsy. Caught before the c sweep ran; $c = 0$ is precisely the setting that turns the correction off and is the consistency check of item 2, so it would have read as “the control variate does nothing”.

The noise floor. Because **random_k** and **xbar_conf** are the same algorithm under **edlm**, their difference measures pure run-to-run noise: hit counts of 30 against 49 on one QED cell. The seed replication puts a number on it directly — see the standard errors above — and it is the reason every comparison here is paired and replicated.

7 Verdict

Confidence-based position selection is **closed as a negative result on the primary metric**, refuted along three independent routes:

1. no arm beats **bernoulli** on hits, with 5 seeds and paired tests;
2. no control-variate coefficient rescues it — $c = 1$ is optimal and still below baseline;
3. no time window separates the quality gain from the diversity cost.

What survives is narrow but real: **arm E raises the mean property of novel molecules significantly, by the largest margin of any arm.** Under an objective that scores mean quality rather than count-above-a-bar, the contribution holds. Under the budgeted-hit-count metric this work adopted, it does not.

The capability argument for **marginal** — that this is definable only there — is intact, but it now buys a capability that does not pay on the metric that matters, so it cannot carry the justification on its own.